

AI-Based Tender Intelligence Platform

Complete Functional & Technical Specifications for Backend & ML Teams

Target Audience: Backend & ML Engineers | Release: v3.2 Production Ready | Date: October 2026

ARCHITECTURAL DIRECTIVE FOR BACKEND & ML TEAMS

This document establishes the end-to-end functional requirements, API contracts, ML pipeline interfaces, database schemas, and verification checklists required to build the AI Procurement & Bid Management Platform. All modules must be implemented strictly following the structured JSON schemas and async lifecycles documented herein.

1. EXECUTIVE PLATFORM OVERVIEW

The AI-Based Tender Intelligence & Bid Management Platform is an enterprise-grade procurement co-pilot designed to automate the extraction, evaluation, compliance auditing, risk profiling, and proposal generation for government and infrastructure bids. It eliminates manual review of 100+ page tender PDFs, evaluates bid feasibility against corporate qualifications in seconds, and provides explainable Go/No-Go decision intelligence.

1.1 Core High-Level System Architecture

- Presentation Layer:** React 19, Tailwind CSS, ShadCN primitives, Recharts analytics, and multi-workspace tender explorer.
- API & Orchestration Layer:** Python FastAPI asynchronous service, JWT RBAC security, multipart PDF upload pipelines.
- AI / ML Pipeline Layer:** Google Gemini 1.5 Flash/Pro, multimodal OCR, text-embedding-004, semantic diff, and RAG knowledge retrieval.
- Data & Vector Storage Layer:** PostgreSQL/SQLite relational database for users/tenders/clauses + FAISS/ChromaDB/pgvector for embeddings.
- Corporate Vault Layer:** Secure document repository tracking licenses, balance sheets, ISO certifications, and auto-matching requirements.

2. MACHINE LEARNING & AI ENGINEERING SPECIFICATIONS

The ML / AI Engineer is responsible for the text parsing pipeline, structured LLM extraction, deterministic eligibility comparison, risk heuristic profiling, semantic vector retrieval, contradiction detection, and generative drafting.

Module ML-1: Document Ingestion & Multimodal OCR Pipeline

- Input:** Tender PDFs (digital vector PDFs and multi-page scanned image documents), Corrigenda notices, and BOQ spreadsheets.
- Processing Pipeline:** Extract raw text via pdfplumber / PyPDF2. Route scanned pages through OCR (Tesseract / pdf2image or Gemini Multimodal).
- Layout Normalization:** Identify sections, preserve clause hierarchies (e.g., Section III Clause 4.2.1), and strip boilerplate headers/footers.

Module ML-2: Structured Entity & Clause Extraction Engine

- Target Schema Extraction:** Call Gemini 1.5 Flash using structured JSON Schema outputs to enforce 100% deterministic schema compliance.
- Core Metadata Extraction:** Title, issuing authority, project value (Cr), EMD amount (Lakhs), submission deadline, and executive summary.
- Clause Categorization:** Automatically segment requirements into 4 distinct categories:
 - * Financial: Minimum average annual turnover, liquid asset requirements, net worth criteria, bank solvency limits.
 - * Experience: Minimum completed project count, single contract threshold value (Cr), years of active corporate operations.
 - * Technical & Machinery: Mandatory equipment inventory (segment launchers, batching plants, pavers) and manpower count.
 - * Documentation: Statutory requirements (Class-A contractor license, ISO 9001/14001, GST, PAN, EPF registrations).
- Confidence Metric:** Calculate extraction confidence score (0.0 to 1.0) per clause and aggregated for the document.

Module ML-3: Explainable Eligibility & Compliance Engine

- Deterministic Rule Matching:** Cross-compare extracted tender thresholds against the active CompanyProfile record.

- **Clause-Level Explainability:** Assign every clause an audit status (PASS, FAIL, WARN) accompanied by an explicit text justification.
- **Equipment Parser:** Extract machinery types and counts; check company assets; differentiate owned vs. permitted lease equipment.
- **Weighted Composite Formula:** Calculate overall eligibility score: $(0.35 \times \text{Financial}) + (0.30 \times \text{Experience}) + (0.20 \times \text{Docs}) + (0.15 \times \text{Tech})$.

Module ML-4: Risk Intelligence & Threat Scoring

- **Penalty Trap Detection:** Scan General and Special Conditions of Contract (GCC/SCC) for punitive liquidated damages (>0.5%/week or cap >10%).
- **Risk Indices Generated:** Calculate Financial Risk, Technical Risk, Compliance Risk, Documentation Risk, and Overall Risk Score (0-100%).
- **Mitigation Strategy Engine:** Auto-generate recommended risk mitigations (e.g., 'Draft 15-day buffer in structural concrete schedule').

Module ML-5: Decision Support & Suitability Engine (Go / No-Go)

- **Suitability Multi-Factor Score:** Synthesize Profitability (30%), Inverted Competition (25%), Inverted Difficulty (25%), and Travel Proximity (20%).
- **Go / No-Go Classification:** Verdict output as 'Go' (>=75%), 'Cautious' (50-74%), or 'No-Go' (<50%) with supporting bullet logic.
- **Bid Readiness Index:** Formula: $0.4 \times \text{Eligibility\%} + 0.6 \times \text{Document Readiness\%}$ (tracks uploaded vs missing checklist items).
- **Dynamic Action Plan:** Generate ordered step-by-step resolution tasks to achieve 100% bid readiness (e.g., 'Step 1: Renew ISO 9001 in Vault').

Module ML-6: Contradiction & Pre-Bid Negotiation Assistant

- **Contradiction Pipeline:** Cross-scan sections to identify internal conflicts (e.g., conflicting completion periods between NIT and Clause 11.4).
- **Pre-Bid Question Generator:** Draft formal clarification letters for official pre-bid queries formatted with clause references.

Module ML-7: Corrigendum / Amendment Semantic Diff Tracker

- **Multi-Document Comparison:** Diff base tender against newly published corrigendum PDFs.
- **Delta Detection:** Flag extensions in deadlines, reductions in EMD fees, and alterations to technical criteria.
- **Impact Re-Evaluation:** Automatically trigger re-scoring (e.g., 'EMD lowered from Rs. 85L to 75L: Status updated to PASS').

Module ML-8: Semantic Vector Index & Similar Tender Retrieval

- **Dense Vector Indexing:** Embed summaries and clauses using text-embedding-004 (or sentence-transformers); store in FAISS / ChromaDB.
- **Historical Similarity:** Perform cosine similarity matching across past won/lost bids to calculate project similarity percentages.
- **Offline Resilience:** Provide local TF-IDF / BM25 fallback engine when external embedding APIs are unavailable.

Module ML-9: Tender RAG Chatbot (Document QA)

- **Retrieval Pipeline:** Chunk document (1000 chars, 200 overlap), execute hybrid keyword+vector search to retrieve top-3 relevant context chunks.
- **Hallucination-Free Generation:** Strict grounding prompt ensuring answers reference only source text with clause/page citations.

Module ML-10: AI Proposal Generator & Bid Synthesizer

- **Template Synthesis:** Synthesize formal bid sections: Technical Qualifications Bid, Methodology Work Plan, Site Safety Plan, and BOQ Sheets.
- **Dynamic Injection:** Autofill company registration numbers, ISO audit dates, turnover figures, and machinery lists into proposal drafts.

3. BACKEND ENGINEERING SPECIFICATIONS

The Backend Engineer is responsible for API routing, security, data persistence, file storage, async task management, and business logic.

Module BE-1: Authentication & Role-Based Access Control (RBAC)

- **Endpoints:** POST /api/auth/register, POST /api/auth/login, GET /api/auth/me.
- **Security:** Password hashing using bcrypt; JWT tokens with configurable expiration (HS256); bearer token middleware.
- **Roles:** Admin (full rights), Company Manager (upload/edit/approve), Estimator (view/chat/draft), Employee (read-only).

Module BE-2: Company Profile & Corporate Vault Management

- **Profile CRUD:** GET /api/profile, PUT /api/profile. Stores turnover, experience years, past projects, equipment, and certifications.
- **Auto-Recalculation Hook:** Updating company profile triggers asynchronous re-evaluation of all active tenders in database.
- **Corporate Document Vault:** File directory tracking licenses and tax records; monitors certificate expiry; auto-matches with tender checklists.

Module BE-3: Tender Lifecycle & File Pipeline

- **Tender Upload:** POST /api/tenders/upload accepts multipart PDF. Saves file, calls ML pipeline, persists clauses/checklists, indexes in vector DB.
- **Catalog CRUD:** GET /api/tenders, GET /api/tenders/{id}, DELETE /api/tenders/{id} (cascades file deletion and vector index purge).
- **Status Tracking:** Maintains lifecycle state: 'pending', 'analyzed', 'error', 'discovery_bookmarked'.

Module BE-4: Tender Discovery & Portal Aggregation

- **Discovery Route:** GET /api/tenders/discovery supports query filtering by keyword (q), department (dept), and value range (min_val, max_val).
- **Bookmarking/Pinning:** PUT /api/tenders/{id}/bookmark toggles pinned status; imports mock/live discovery tender into active database.
- **Saved Bids:** GET /api/tenders/saved returns filtered bookmarked tenders.

Module BE-5: Corrigendum Versioning & Deadline Auto-Sync

- **Amendment Ingestion:** POST /api/tenders/{id}/amendment accepts corrigendum PDF, runs diff engine, stores amendment record.
- **Dynamic Field Updates:** Regex parsers extract extended deadlines and revised EMD values to automatically update parent tender records.

Module BE-6: Checklist & Bid Readiness Tracking

- **Checklist Updates:** PUT /api/tenders/{id}/checklist/{item_id} updates document status (missing, uploaded, vault_matched).
- **Score Re-computation:** Checklist status change automatically updates bid_readiness_score and recalculates the dynamic action plan.

Module BE-7: Team Collaboration & Activity Audit Trail

- **Task Management:** Create and toggle status of bid preparation tasks (todo, in_progress, completed) linked to specific tenders.
- **Activity Audit:** Log all critical events: document uploads, eligibility evaluations, corrigendum diffs, and checklist modifications.

Module BE-8: Analytics & Aggregations

- **Pipeline Valuation:** Aggregate active bid pipeline values, average risk distribution, and readiness percentages for executive reporting.

4. DATABASE SCHEMA & DATA MODELS

The database architecture requires 8 primary relational entities supporting cascading integrity:

Table Name	Key Columns	Relationships / Foreign Keys	Description
users	id, username, email, password_hash, role	One-to-Many with team tasks	User credentials & RBAC authorization
company_profiles	id, company_name, turnover, equipment, ..	Root enterprise entity	Bidder corporate qualifications pro..
tenders	id, title, value, EMD, deadline, status..	1:M clauses, amendments, che..	Primary tender records with AI scores
tender_clauses	id, tender_id, category, clause_text, s..	FK -> tenders.id (cascade de..	Extracted clauses with PASS/FAIL st..
tender_amendments	id, tender_id, title, changes_summary, ..	FK -> tenders.id (cascade de..	Corrigendum documents & diff logs
document_checklists	id, tender_id, document_name, status, v..	FK -> tenders.id (cascade de..	Mandatory submission files & vault ..
vault_documents	id, document_name, category, file_path,..	Corporate asset repository	Reusable credentials & expiration d..
team_tasks	id, tender_id, title, assignee, status	FK -> tenders.id (cascade de..	Collaborative bid prep tasks

5. COMPLETE REST API ENDPOINT MATRIX

Method	Endpoint Route	Function / Payload	Owner
POST	/api/auth/register	User signup (username, email, password, role)	Backend
POST	/api/auth/login	User login, issues JWT access_token	Backend
GET	/api/auth/me	Fetch current authenticated user context	Backend
GET	/api/profile	Retrieve company profile parameters	Backend
PUT	/api/profile	Update profile & trigger tender re-scoring	Backend + ML
GET	/api/tenders	List all analyzed tenders with risk & match scores	Backend
POST	/api/tenders/upload	Upload PDF -> triggers OCR, extraction & audit	Backend + ML
GET	/api/tenders/{id}	Full tender details, clauses, risks, checklists	Backend
DELETE	/api/tenders/{id}	Delete tender record, PDF file & vector embeddings	Backend
GET	/api/tenders/discovery	Query open tenders with keyword/dept/value filters	Backend
PUT	/api/tenders/{id}/bookmark	Toggle pinned/saved state in catalog	Backend
GET	/api/tenders/saved	Fetch all bookmarked/pinned tenders	Backend
POST	/api/tenders/{id}/chat	RAG Q&A over tender specifications	ML Engine
POST	/api/tenders/{id}/amendment	Upload corrigendum PDF & generate diff log	Backend + ML
PUT	/api/tenders/{id}/checklist/{item}	Update document upload/vault status	Backend
GET	/api/tenders/{id}/similar	Retrieve top-k similar historical tenders	ML Engine

6. PLATFORM CAPABILITIES SUMMARY

The AI-Based Tender Intelligence Platform delivers end-to-end automation across the bidding lifecycle through 6 core pillars:

- 1. **Automated Ingestion & Parsing**: Converts multi-hundred page tender PDFs into clean structured data in seconds, replacing tedious manual reading.
- 2. **Explainable Eligibility Scoring**: Compares company turnover, machinery, and past contracts against tender rules with clause-by-clause PASS/FAIL justifications.
- 3. **Risk Intelligence & Threat Auditing**: Detects penalty clauses, defect liability traps, and liquidity blockages before management commits resources.
- 4. **Decision Support & Action Plans**: Computes Suitability Indices, issues definitive Go/No-Go recommendations, and generates step-by-step deficiency checklists.
- 5. **Corrigenda & Contradiction Tracking**: Flags internal document contradictions and auto-diffs amendments to prevent missing revised deadlines or lower EMDs.
- 6. **RAG Q&A & Proposal Synthesis**: Empowers bid managers to query specifications via conversational AI and auto-draft qualification proposals.

7. ENGINEERING IMPLEMENTATION CHECKLIST POINTS

Use the following actionable checklist points to track development progress across teams:

7.1 Machine Learning & AI Engineering Checklist

- ☐ ML-01 **PDF Ingestion & OCR Pipeline**
Implement dual-mode parser: direct text extraction via pdfplumber + fallback OCR via pytesseract/pdf2image for scanned documents.
- ☐ ML-02 **Deterministic Schema Extractor**
Configure Gemini 1.5 Flash with strict JSON Schema output mode to reliably extract title, value, EMD, deadlines, clauses, and risks.
- ☐ ML-03 **Extraction Confidence Scorer**
Implement extraction confidence score computation (0.0 to 1.0) based on token probability or model self-evaluation.
- ☐ ML-04 **Clause Classification System**
Automatically classify extracted clauses into Financial, Experience, Technical/Machinery, and Documentation categories.
- ☐ ML-05 **Deterministic Eligibility Matcher**
Build comparison logic matching company turnover, max completed project value, and experience years against criteria.

- ☐ **ML-06 Equipment & Machinery Matcher**
Implement regex & NLP parser to match required plant machinery against company asset inventory (checking quantity and owned vs leased).
- ☐ **ML-07 Weighted Scoring Engine**
Implement composite eligibility formula (35% Financial, 30% Experience, 20% Documentation, 15% Technical).
- ☐ **ML-08 Suitability & Go/No-Go Engine**
Implement Suitability rating (profitability, competition, difficulty, distance) and Go/No-Go decision logic with rationale.
- ☐ **ML-09 Dynamic Action Plan Generator**
Generate prioritized step-by-step action items dynamically derived from failed clauses and missing checklist documents.
- ☐ **ML-10 Contradiction & Pre-Bid Assistant**
Develop cross-clause contradiction detector and generate structured pre-bid clarification questionnaires.
- ☐ **ML-11 Corrigendum Diff Engine**
Implement semantic diff between original tender and addendum PDFs; extract revised submission deadlines and revised EMDs.
- ☐ **ML-12 Vector Store & RAG Chatbot**
Set up text-embedding-004 vector indexing, hybrid top-k chunk retrieval, cosine similarity search, and cited RAG answers.

7.2 Backend Engineering Checklist

- ☐ **BE-01 Database Schema & Migrations**
Set up SQLAlchemy models (Users, CompanyProfile, Tenders, Clauses, Amendments, Checklists, Vault, Tasks) with Alembic migrations.
- ☐ **BE-02 Authentication & Security**
Implement bcrypt password hashing, JWT access token generation, authentication middleware, and RBAC role validation.
- ☐ **BE-03 Company Profile Endpoints**
Build GET and PUT /api/profile endpoints; attach async event hook to recalculate eligibility across active tenders on update.
- ☐ **BE-04 Multipart Tender Upload API**
Implement POST /api/tenders/upload; handle file storage, async parser queuing, and atomic DB transaction commits.
- ☐ **BE-05 Tender Catalog & Cascade Delete**
Implement GET /api/tenders, GET /api/tenders/{id}, and DELETE /api/tenders/{id} with cascade deletion of files and vectors.
- ☐ **BE-06 Tender Discovery & Search**
Build GET /api/tenders/discovery with keyword search, department filtering, and contract value range filters.
- ☐ **BE-07 Bookmark / Pinning Logic**
Implement PUT /api/tenders/{id}/bookmark; auto-import discovery placeholder tenders into the permanent catalog.
- ☐ **BE-08 Corrigendum Ingestion API**
Build POST /api/tenders/{id}/amendment; trigger ML diff engine and automatically update parent tender deadline and EMD fields.
- ☐ **BE-09 Document Checklist Pipeline**
Implement PUT /api/tenders/{id}/checklist/{item_id}; re-calculate bid_readiness_score upon status updates.
- ☐ **BE-10 Vault Matching Automation**
Cross-check checklist requirements against vault documents; auto-set status to 'vault_matched' when valid certificates exist.
- ☐ **BE-11 Team Collaboration Tasks**
Implement task creation, assignment, and status toggle endpoints; build immutable audit log for tender actions.
- ☐ **BE-12 CORS, Validation & Docs**
Configure CORS middleware for frontend origins, Pydantic v2 request/response validation, and Swagger UI at /docs.

7.3 DevOps, Infrastructure & Testing Checklist

- ☐ **OPS-01 Environment Configuration**
Set up .env configuration with GEMINI_API_KEY, SECRET_KEY, DATABASE_URL, and UPLOAD_DIR settings.

- ☐ **OPS-02 Background Worker Queue**
Configure Celery/Redis or FastAPI BackgroundTasks for asynchronous long-running PDF parsing jobs.
- ☐ **OPS-03 Automated Unit Tests**
Write automated tests asserting eligibility score calculations, Go/No-Go branch logic, and action plan generation.
- ☐ **OPS-04 API Integration Tests**
Implement pytest suite testing Auth lifecycle, tender upload, checklist updates, and RAG chat endpoints.
- ☐ **OPS-05 RAG Accuracy & Benchmark**
Establish a precision/recall evaluation script over 10 sample tender queries to verify citation accuracy.
- ☐ **OPS-06 Production Deployment**
Prepare Dockerfile multi-stage builds and docker-compose configurations for FastAPI, PostgreSQL, and Redis.